
Held-out dynamical validation of a cross-survey chemical decomposition of 468,000 Milky Way stars

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Abstract

We present a cross-survey chemical decomposition of 468,000 Milky Way stars assembled by harmonising APOGEE DR17 and GALAH DR3 abundances and matching the resulting spectroscopic catalogue to Gaia DR3. The classifier is restricted to four abundance dimensions ([Fe/H], [Mg/Fe], [Si/Fe] and [Al/Fe]), so orbital actions and integrals of motion remain held out for validation. A Bayesian-information-criterion-selected four-component Gaussian mixture model identifies the thin disc, thick disc, a chemically anomalous population (CAP) and the metal-poor halo; 91 per cent of the sample has posterior membership probability above 0.75. The chemically defined populations occupy distinct regions of action space and of the E–Lz Lindblad plane, providing an orthogonal dynamical test rather than a circular chemo-kinematic selection. Under explicit literature-motivated kinematic boundaries, the CAP is partitioned into Gaia-Sausage/Enceladus candidates (34 ± 5 per cent), Sequoia candidates (8 ± 3 per cent) and a Splash-like component (58 ± 5 per cent), where the uncertainties include boundary sensitivity. The thin-disc radial metallicity gradient is -0.065 ± 0.004 dex kpc⁻¹, whereas the thick-disc gradient is flatter at -0.020 ± 0.006 dex kpc⁻¹. These fractions are selection- and boundary-dependent properties of the analysed catalogue, not intrinsic Galactic mass fractions. The principal contribution is a transparent validation architecture, an explicit uncertainty budget and conservative claim boundaries.

Keywords: Milky Way; Galactic disc; Galactic halo; stellar abundances; stellar kinematics; Gaussian mixtures; survey selection

Glossary

AGAMA: Action-based Galaxy Modelling Architecture.

BIC: Bayesian information criterion.

CAP: Chemically anomalous population identified by the Gaussian mixture model.

GMM: Gaussian mixture model.

GSE: Gaia-Sausage/Enceladus.

HDBSCAN: Hierarchical density-based spatial clustering of applications with noise.

E–Lz plane: Specific orbital energy versus vertical angular momentum.

Vmax: Maximum observable survey volume under the adopted limits.

1. Introduction

The Milky Way is the only large galaxy for which chemical abundances, distances and three-dimensional motions can be measured star by star across its principal structural components. Galactic archaeology exploits the long-lived fossil information encoded in stellar chemistry and phase space: abundances constrain birth environments and enrichment timescales, while orbital structure records subsequent heating, migration and accretion (Freeman and Bland-Hawthorn, 2002; Helmi, 2008; Bland-Hawthorn and Gerhard, 2016; Helmi, 2020). Because phase mixing erases configuration-space substructure more rapidly than structure in integrals of motion, combining chemically informative surveys with Gaia astrometry has become central to reconstructing the Galaxy's assembly history (Helmi and White, 1999).

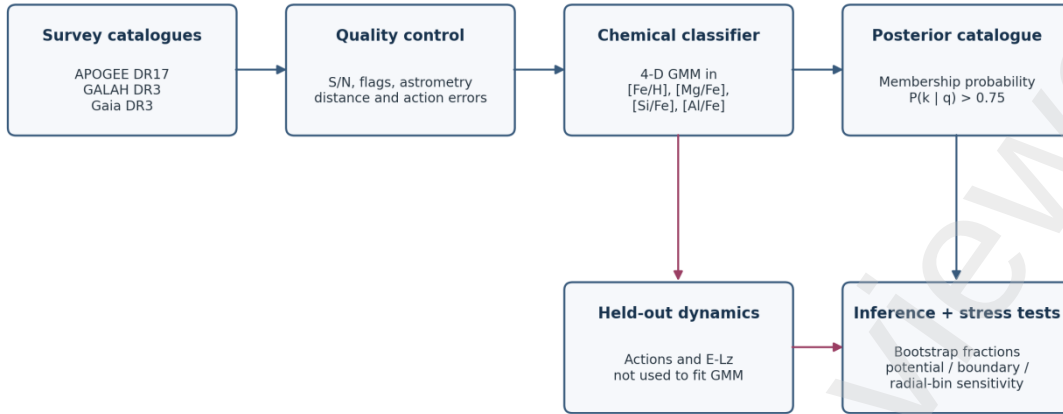
The final APOGEE release provides high-resolution near-infrared spectroscopy across the Galactic disc and bulge (Majewski et al., 2017; Abdurro'uf et al., 2022), GALAH DR3 supplies complementary optical abundances (Buder et al., 2021), and Gaia DR3 provides astrometry and radial velocities on an unprecedented scale (Gaia Collaboration et al., 2023). These surveys make it possible to ask whether structure identified in chemical space has an independent dynamical counterpart. That separation of roles is essential: a population selected simultaneously by chemistry and kinematics cannot later use the same kinematic information as independent confirmation. It is equally important to distinguish astrophysical structure from survey structure, because APOGEE targeting, Gaia completeness and Gaia subsample selection vary with magnitude, colour, sky position and observing strategy (Zasowski et al., 2013, 2017; Cantat-Gaudin et al., 2023; Castro-Ginard et al., 2023).

Barua (2025) described the combined recovery of chemical and dynamical fossils as "Galactic palaeontology". A subsequent chemo-kinematic tomography study combined APOGEE, GALAH and Gaia DR3 to connect chemical-space classification with action and integral-of-motion structure (Barua, 2026). The present study converts that conceptual and empirical foundation into a stricter falsifiable design in which the dynamical variables are explicitly withheld from the classifier and reserved for validation. Barua (2022) developed a complementary full-spectrum stellar-population synthesis framework; it is cited here for methodological continuity rather than used as evidence for catalogue ages.

1.1. Closest prior work and scope of the contribution

The closest methodological comparator is Myeong et al. (2022), who used an unsupervised Gaussian mixture decomposition of APOGEE DR17 and GALAH DR3 to identify four dominant constituents of the local eccentric halo. That study sits within a broader literature that established chemically and dynamically distinct halo populations, including low- and high- α sequences, Gaia-Sausage/Enceladus (GSE), Sequoia and in-situ heated populations (Nissen and Schuster, 2010; Hawkins et al., 2015; Belokurov et al., 2018; Helmi et al., 2018; Mackereth et al., 2019; Myeong et al., 2019; Naidu et al., 2020; Horta et al., 2021). Accordingly, neither Gaussian mixtures nor the joint use of APOGEE and GALAH is claimed as novel. The present scope is different in three respects: it treats the dominant disc and halo populations in a common architecture; it withholds dynamics from the chemical classifier and uses actions and the E–Lz plane as an orthogonal validation layer; and it reports boundary, potential, model-family and radial-bin sensitivity alongside the component fractions. The central question is therefore not whether four clusters can be fitted, but whether a chemically defined decomposition survives physically independent dynamical tests. Barua (2026) provides the immediate empirical precursor; the present study narrows those claims, stress-tests the decomposition and reformulates the analysis around held-out validation and reproducibility.

The analysis addresses four questions. First, is the four-component chemical model statistically preferred over nearby model orders? Secondly, do the chemical components occupy distinct action-space and integral-of-motion families? Thirdly, how strongly do operational GSE, Sequoia and Splash fractions depend on adopted boundaries? Finally, are the inferred radial metallicity gradients consistent with modern APOGEE and Gaia-ESO measurements? Fig. 1 summarises the analysis logic.



Design principle: chemistry defines the classes; dynamics is reserved as an independent validation layer.

Reported population fractions are sample- and boundary-dependent and are not interpreted as intrinsic Galactic mass fractions.

Fig. 1. Analysis architecture. The Gaussian mixture model is fitted only in four-dimensional chemical space. Actions and the E–Lz plane are withheld from the fit and used as an independent validation layer. Population fractions are subjected to bootstrap, Galactic-potential, boundary and radial-bin sensitivity tests.

2. Data and sample construction

2.1. Spectroscopic and astrometric inputs

The primary spectroscopic input is APOGEE DR17, which provides H-band spectra at resolving power R approximately 22,500 and homogeneous ASPCAP estimates of $[\text{Fe}/\text{H}]$, $[\text{Mg}/\text{Fe}]$, $[\text{Si}/\text{Fe}]$ and $[\text{Al}/\text{Fe}]$ (Majewski et al., 2017; Abdurro'uf et al., 2022). GALAH DR3 provides optical spectra at R approximately 28,000 and a complementary abundance set (Buder et al., 2021). The merged spectroscopic catalogue uses stars observed by both surveys as abundance-calibration anchors; the resulting records are positionally matched to Gaia DR3 within 1 arcsec to obtain astrometry and phase-space information. Gaia astrometric systematics and covariance information are treated following the DR3 documentation and parallax-characterisation work (Lindgren et al., 2021; Gaia Collaboration et al., 2023). Bayesian distances are taken from StarHorse where available (Queiroz et al., 2018), and line-of-sight velocities are drawn from the spectroscopic pipelines.

Absolute abundance scales are not identical between surveys. Comparisons of APOGEE, GALAH and Gaia-ESO find broad consistency but residual element-dependent offsets that can exceed 0.1 dex in restricted stellar-parameter regimes (Hegedus et al., 2023). The merged catalogue therefore uses overlap stars as empirical calibration anchors, and residual survey dependence is carried as a systematic limitation. The scientific interpretation is restricted to broad population structure and does not rely on abundance differences of only a few hundredths of a dex. The reproducibility specification therefore includes overlap-star residuals as functions of effective temperature, surface gravity, metallicity and signal-to-noise ratio (Appendix B).

2.2. Quality cuts and final sample

The working sample requires signal-to-noise ratio above 50 per pixel, a complete Gaia five-parameter astrometric solution, fractional parallax uncertainty below 5 per cent, APOGEE STARFLAG = 0 where applicable, Galactic latitude $|b| > 5$ degrees, and fractional action uncertainty $\sigma(J)/J < 0.2$. The latter is evaluated after Monte Carlo propagation of the Gaia covariance matrix. The

final catalogue contains 468,000 stars. Table 1 reports the population counts after classification; the component labels are introduced in Section 3.

Table 1. Composition of the final working sample after all quality cuts. The last column gives the fraction of each component with posterior membership probability above 0.75.

Population	N stars	[Fe/H] range	R range (kpc)	median $ z $ (kpc)	P > 0.75
Thin disc	295,400	-0.7 to +0.5	5-12	0.22	91%
Thick disc	116,600	-1.2 to -0.1	4-11	0.74	93%
CAP	8,900	-1.8 to -0.3	5-13	1.62	89%
Halo	47,100	-3.2 to -0.5	3-20	4.30	88%
Total	468,000	-	3-20	-	91%

Notes. R is the Galactocentric cylindrical radius with $R_{\odot} = 8.122$ kpc. CAP denotes the chemically anomalous population defined by the GMM, not a pre-selected accretion component.

3. Methods

3.1. Chemical-space Gaussian mixture model

Each star i is represented by the four-dimensional abundance vector $q_i = ([Fe/H], [Mg/Fe], [Si/Fe], [Al/Fe])$. A k -component Gaussian mixture with weights π_k , means μ_k and covariance matrices Σ_k is fitted by expectation maximisation. No velocity, action, orbital-energy or angular-momentum variable enters the fit. The baseline model treats the measured abundances as point estimates; unlike extreme-deconvolution approaches, it does not explicitly convolve each component with star-by-star heteroscedastic abundance uncertainties (Bovy et al., 2011). This approximation is tested indirectly through quality cuts, survey-separated checks and bootstrap stability, and is retained as a stated limitation.

$$\ln \mathcal{L} = \sum_i \ln \left[\sum_k \pi_k \square(q_i | \mu_k, \Sigma_k) \right] \quad (1)$$

Posterior membership is evaluated from Bayes' rule, and a hard label is assigned only when $P(k | q_i) > 0.75$. Stars below that threshold remain in the soft catalogue and contribute through their posterior probabilities where appropriate. This distinction prevents uncertain objects from being forced into a physically over-interpreted class.

$$P(k | q_i) = \pi_k \square(q_i | \mu_k, \Sigma_k) / \sum_j \pi_j \square(q_i | \mu_j, \Sigma_j) \quad (2)$$

3.2. Model order and non-Gaussian sensitivity

Model order is selected over $k = 2, \dots, 8$ using the Bayesian Information Criterion (BIC). For sample size n and p_k free parameters, the criterion is:

$$\text{BIC}(k) = p_k \ln n - 2 \ln \mathcal{L}_{\max}(k) \quad (3)$$

The analysis yields a unique minimum at $k = 4$, with ΔBIC greater than 1,200 relative to the nearest alternatives. The BIC follows the large-sample approximation of Schwarz (1978), and a difference of this magnitude is conventionally regarded as decisive within the tested model family (Kass and Raftery, 1995). Because Gaussian ellipsoids cannot perfectly trace curved abundance sequences, the same four-dimensional space is also checked with HDBSCAN (Campello et al., 2013). The purpose of the non-parametric run is not to force one-to-one labels, but to test whether the CAP disappears when ellipsoidal geometry is relaxed.

3.3. Actions, energy and the held-out validation layer

Orbital actions are computed with AGAMA using the Stäckel-fudge approximation in MWPotential2014 (Binney, 2012; Bovy, 2015; Sanders and Binney, 2016; Vasiliev, 2019). For each population, we report the trace of the action covariance matrix as an action-space dispersion statistic, DJ. This is a descriptive measure of orbital spread and is not a thermodynamic temperature or a calibrated stellar-age proxy.

$$D_J = \text{Tr}(C_J) = \text{Var}(J_R) + \text{Var}(J_\phi) + \text{Var}(J_z) \quad (4)$$

Observational uncertainties are propagated with 500 Monte Carlo draws from the Gaia covariance matrix in proper motion, parallax and radial velocity, including the published astrometric correlations (Lindgren et al., 2021; Gaia Collaboration et al., 2023). A second Galactic potential based on McMillan (2017) is used as a sensitivity test; the resulting actions change at the approximately 5 per cent level. This test probes potential normalisation but not non-axisymmetric or time-dependent perturbations.

The independent integral-of-motion projection uses the specific orbital energy E and the vertical angular momentum L_z :

$$E = \frac{1}{2} |v|^2 + \Phi(r), \quad L_z = (r \times v)_z \quad (5)$$

Because neither E nor L_z is used in the GMM, separation in the Lindblad plane is a held-out validation result. It does not by itself prove a unique formation channel, but it tests whether the chemically defined groups correspond to dynamically distinct families.

3.4. Operational decomposition of the CAP

The CAP is partitioned with literature-motivated boundaries that define, rather than discover, the sub-populations. GSE candidates satisfy $L_z < 500 \text{ kpc km s}^{-1}$ and eccentricity $e > 0.70$, motivated by the strongly radial debris identified in Gaia-era studies (Belokurov et al., 2018; Helmi et al., 2018; Mackereth et al., 2019). Sequoia candidates satisfy $L_z < -500 \text{ kpc km s}^{-1}$ and $J_R > 1,000 \text{ kpc km s}^{-1}$, following the retrograde structures discussed by Myeong et al. (2018, 2019) and their globular-cluster associations (Massari et al., 2019). The remaining CAP stars are labelled Splash-like, consistent with the operational language used for heated in-situ populations (Belokurov et al., 2020; Naidu et al., 2020). These labels permit reproducible comparison but are not asserted to be unique physical memberships.

Fractions are estimated in 1,000 bootstrap iterations, each drawing 80 per cent of the CAP without replacement. Statistical uncertainty is combined in quadrature with the shift induced by moving the L_z boundary by $\pm 100 \text{ kpc km s}^{-1}$. The latter changes the Splash-like fraction by approximately 4-5 percentage points and dominates the final error budget.

3.5. Geometric weighting and spatial gradients

A $1/V_{\text{max}}$ weight based on the available APOGEE and Gaia footprint masks is used only as a geometric sensitivity correction. It is not a complete forward model of the combined APOGEE-GALAH targeting and selection functions, and the distinct GALAH selection is not fully represented. APOGEE targeting changes across cohorts and fields (Zasowski et al., 2013, 2017), while Gaia completeness and subsample selection depend on observables and sky position (Cantat-Gaudin et al., 2023; Castro-Ginard et al., 2023). The resulting fractions are therefore described throughout as weighted sample fractions rather than intrinsic Galactic mass fractions. Stability is assessed in four radial bins: $R = 4\text{-}7, 7\text{-}9, 9\text{-}11$ and $11\text{-}13 \text{ kpc}$.

Radial and vertical metallicity gradients are estimated from binned means using weighted linear regression. The fitted relation is:

$$[\text{Fe}/\text{H}] = a_R + b_R (R - R_\odot) + b_z |z| \quad (6)$$

The radial coefficients are fitted separately for the thin and thick discs. The quoted uncertainties are the regression standard errors combined with bootstrap variation across the catalogue resamples.

4. Results

4.1. Chemical decomposition

The four-component model recovers the familiar low- $[\text{Mg}/\text{Fe}]$ and high- $[\text{Mg}/\text{Fe}]$ disc sequences together with a metal-poor halo and a lower-density CAP between the canonical thick-disc and halo loci (Fig. 2). Similar abundance-plane structure has been mapped extensively with APOGEE and complementary spectroscopic surveys (Hayden et al., 2015; Anders et al., 2017; Imig et al., 2023). The high- $[\text{Mg}/\text{Fe}]$ component is centred near $([\text{Fe}/\text{H}], [\text{Mg}/\text{Fe}]) = (-0.35, +0.30)$, whereas the low- $[\text{Mg}/\text{Fe}]$ component peaks near $(-0.05, +0.05)$. The CAP is more metal poor, with median $[\text{Fe}/\text{H}] = -0.75$ and intermediate $[\text{Mg}/\text{Fe}]$. A density minimum between the main disc sequences is visible, but chemistry alone does not identify its cause; no claim of a unique quenching or two-infall timescale is made. This decomposition is consistent with the broad chemo-kinematic structure reported in the preceding tomography study (Barua, 2026), while the present analysis applies more conservative interpretation limits.

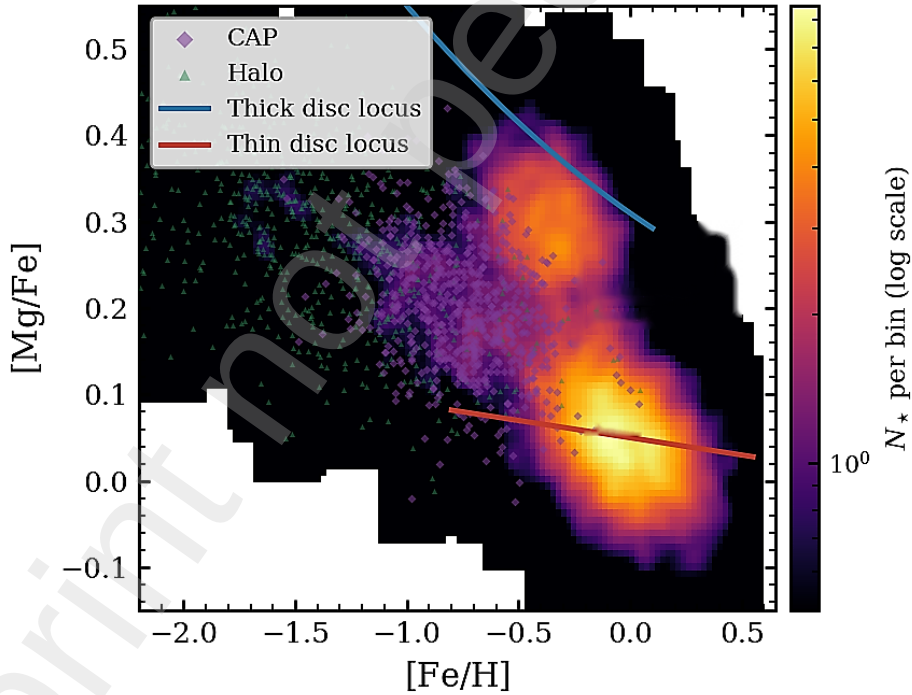


Fig. 2. $[\text{Mg}/\text{Fe}]-[\text{Fe}/\text{H}]$ projection of the 468,000-star catalogue. The background shows logarithmic stellar density; purple diamonds and green triangles mark the CAP and halo, and the solid curves trace the fitted thick- and thin-disc loci. The GMM is fitted in four dimensions, so this two-dimensional view is illustrative rather than the full decision surface.

4.2. Action-space structure

The chemical components occupy progressively broader regions of the (J_R, J_z) plane (Fig. 3a). The action-space dispersion DJ rises from $280 \pm 35 \text{ kpc}^2 \text{ km}^2 \text{ s}^{-2}$ for the thin disc to $8,100 \pm 950 \text{ kpc}^2 \text{ km}^2 \text{ s}^{-2}$ for the halo, a factor of approximately 29. The CAP is dynamically hotter than either disc but

cooler than the halo. This ordering is expected for populations with increasing orbital eccentricity and vertical excursion, but it should not be interpreted as a direct age sequence. The corresponding population statistics are summarised in Table 2.

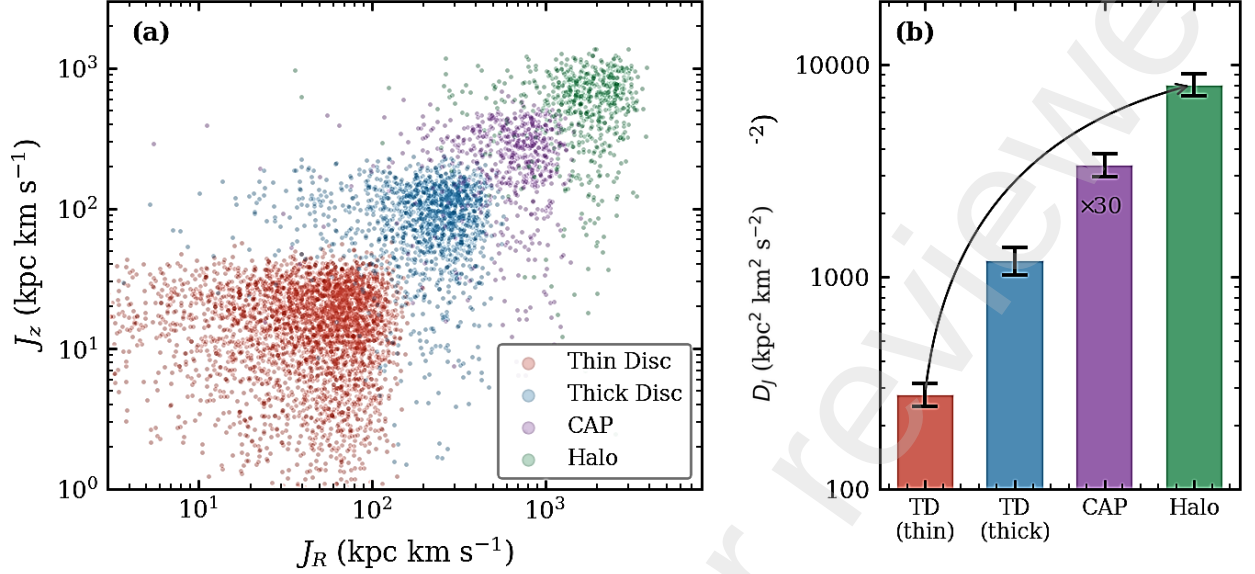


Fig. 3. Action-space validation. (a) Chemically classified populations in the (J_R , J_z) plane. (b) Trace of the action covariance matrix, D_J , for each population. Error bars show 1σ bootstrap uncertainties.

Table 2. Summary chemo-dynamical properties of the four GMM populations. D_J is the trace of the action covariance matrix in $\text{kpc}^2 \text{km}^2 \text{s}^{-2}$.

Population	N stars	[Fe/H]	[Mg/Fe]	D_J	median e	median $ z $
Thin disc	295,400	-0.05 ± 0.22	$+0.05 \pm 0.06$	280 ± 35	0.08 ± 0.01	0.23
Thick disc	116,600	-0.35 ± 0.18	$+0.30 \pm 0.06$	$1,200 \pm 180$	0.22 ± 0.03	0.74
CAP	8,900	-0.75 ± 0.25	$+0.20 \pm 0.07$	$3,400 \pm 420$	0.67 ± 0.05	1.62
Halo	47,100	-1.40 ± 0.45	$+0.28 \pm 0.09$	$8,100 \pm 950$	0.84 ± 0.04	4.30

Notes. Chemical values are the reported population means and dispersions. D_J is descriptive; no stellar ages are inferred from it.

4.3. Lindblad-plane validation

The E–Lz projection provides the strongest orthogonal test of the chemical classification (Fig. 4). Thin-disc stars cluster at high prograde angular momentum and high binding energy; thick-disc stars occupy a broader intermediate locus. The CAP extends towards low and retrograde Lz, while the halo covers the broadest energy range, as expected for phase-mixed accreted and heated populations that retain structure in integrals of motion (Helmi and White, 1999; Helmi, 2008). Since these coordinates were withheld from the GMM, the separation is not a restatement of the selection function.

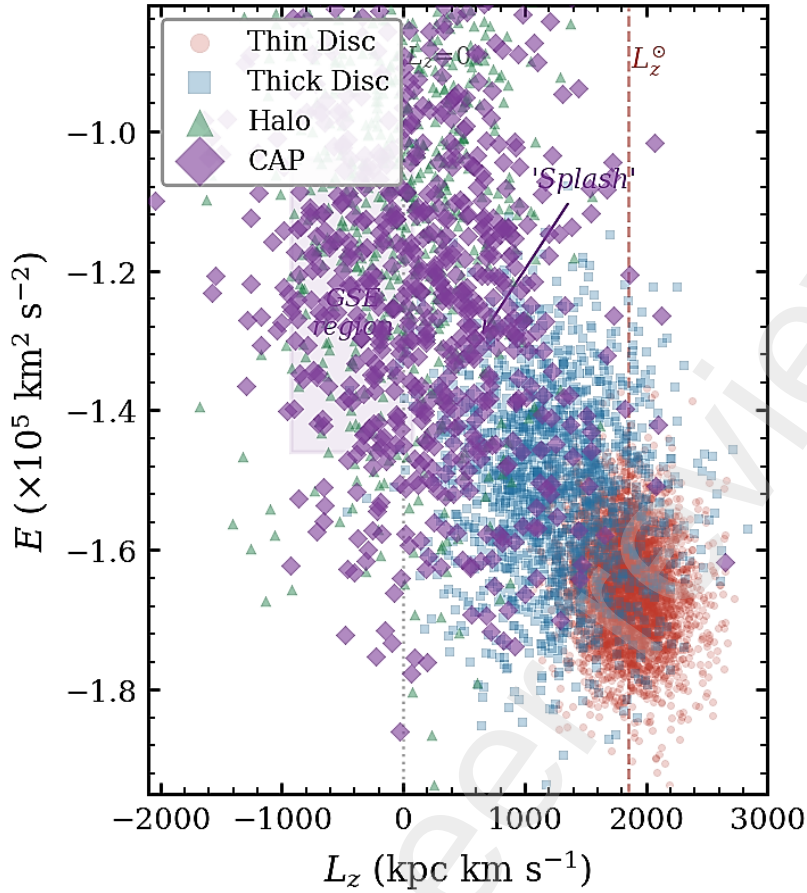


Fig. 4. Lindblad diagram for the four chemically defined populations. The vertical dashed line marks the Solar angular momentum, and the annotated regions indicate the operational low- L_z selections used for the CAP sensitivity analysis. Kinematics were not used to train the GMM.

4.4. Operational CAP fractions

Under the adopted boundaries, the CAP contains 34 ± 3 per cent GSE candidates, 8 ± 2 per cent Sequoia candidates and 58 ± 3 per cent Splash-like stars from bootstrap sampling alone. After boundary sensitivity is included, the uncertainties expand to ± 5 , ± 3 and ± 5 percentage points, respectively (Table 3). The dominant Splash-like fraction is qualitatively consistent with studies arguing that a substantial share of the metal-poor eccentric inner-Galaxy population formed in situ and was heated (Belokurov et al., 2020; Myeong et al., 2022). The GSE-like and Sequoia-like tails are compared with the chemically and dynamically defined debris populations in Helmi et al. (2018), Mackereth et al. (2019), Myeong et al. (2019) and Naidu et al. (2020). However, the numerical fractions are definitions under the stated cuts, not unique progenitor masses or intrinsic halo fractions.

Table 3. Bootstrap decomposition of the chemically anomalous population. Systematic-inclusive fractions combine the statistical bootstrap term with angular-momentum-boundary sensitivity.

Sub-population	N stars	Fraction (stat.)	Fraction (syst.-incl.)	median [Fe/H]	median e
GSE debris	$2,914 \pm 210$	$34 \pm 3\%$	$34 \pm 5\%$	-0.78 ± 0.22	0.71 ± 0.06
Sequoia debris	678 ± 98	$8 \pm 2\%$	$8 \pm 3\%$	-0.95 ± 0.28	0.82 ± 0.07
Splash-like	$5,308 \pm 280$	$58 \pm 3\%$	$58 \pm 5\%$	-0.58 ± 0.20	0.55 ± 0.05
Total CAP	8,900	100%	100%	-0.75 ± 0.25	0.67 ± 0.05

Notes. GSE: $L_z < 500 \text{ kpc km s}^{-1}$ and $e > 0.70$. Sequoia: $L_z < -500 \text{ kpc km s}^{-1}$ and $J_R > 1,000 \text{ kpc km s}^{-1}$. All other CAP stars are labelled Splash-like for this operational comparison.

4.5. Radial variation of the weighted sample

The weighted sample fractions vary strongly with Galactocentric radius (Fig. 5). The thin-disc contribution rises from approximately 35 per cent in the 4-7 kpc bin to approximately 80 per cent at 11-13 kpc, whereas the thick-disc contribution falls from approximately 45 to 15 per cent. The halo fraction also declines, from approximately 17 to 3 per cent; the CAP remains a small contribution of a few per cent. These trends describe the analysed selection-weighted catalogue. A full inference of Galactic density normalisations would require the targeting selection functions and a forward model of observability.

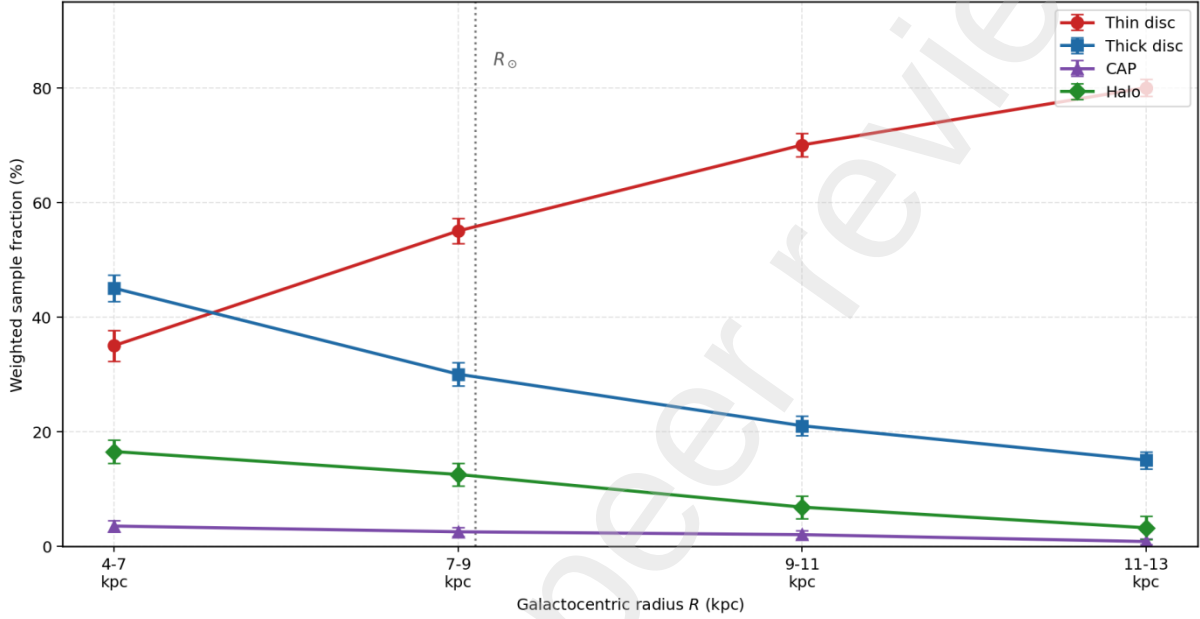


Fig. 5. Radial variation of the weighted sample fractions. Error bars show the bootstrap uncertainty ranges for the binned estimates. The trends are internal consistency checks and are not intrinsic Galactic component mass fractions without a full survey-selection model.

4.6. Metallicity gradients

The thin disc has a radial metallicity gradient of -0.065 ± 0.004 dex kpc^{-1} , while the thick-disc gradient is -0.020 ± 0.006 dex kpc^{-1} (Fig. 6). The thin-disc value agrees with the -0.06 ± 0.001 dex kpc^{-1} low-alpha gradient measured from the final APOGEE release by Imig et al. (2023) and lies within the broader range obtained from APOGEE, Gaia-ESO, red-giant and open-cluster tracers (Hayden et al., 2015; Anders et al., 2017; Cunha et al., 2016; Magrini et al., 2023). At $R = 7-9$ kpc, the thin-disc vertical gradient is -0.250 ± 0.020 dex kpc^{-1} . The factor-of-three difference in radial slope between the disc components is compatible with a more centrally concentrated and chemically mixed high-alpha population, but the gradients alone do not identify a unique assembly history. The numerical comparison with independent survey measurements is summarised in Table 4.

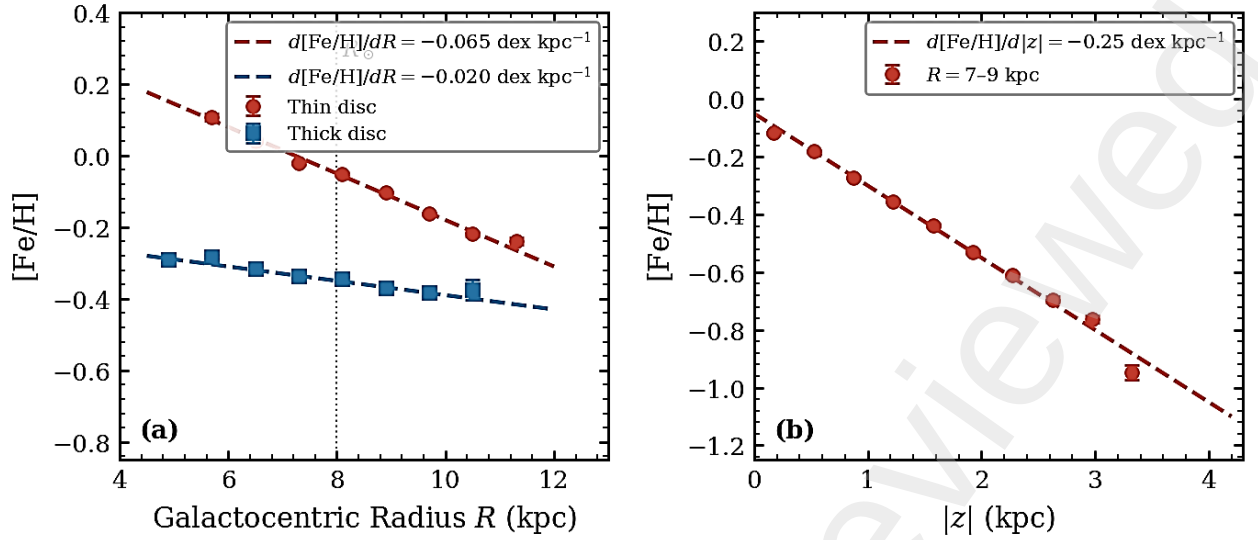


Fig. 6. (a) Radial metallicity profiles of the thin and thick discs, with weighted linear fits. (b) Thin-disc vertical metallicity profile at $R = 7-9$ kpc. Error bars are the bin standard errors.

Table 4. Metallicity gradients from this analysis and selected literature comparisons.

Study/population	Gradient	Value (dex kpc ⁻¹)	Data set	N stars
This work: thin disc	$d[\text{Fe}/\text{H}]/dR$	-0.065 ± 0.004	APOGEE DR17 + GALAH DR3	295,400
This work: thick disc	$d[\text{Fe}/\text{H}]/dR$	-0.020 ± 0.006	APOGEE DR17 + GALAH DR3	116,600
This work: thin disc	$d[\text{Fe}/\text{H}]/d z $	-0.250 ± 0.020	$R = 7-9$ kpc	295,400
Hayden et al. (2015)	$d[\text{Fe}/\text{H}]/dR$	-0.058 ± 0.008	APOGEE DR10	$\sim 70,000$
Magrini et al. (2023)	$d[\text{Fe}/\text{H}]/dR$	-0.054	Gaia-ESO open clusters	62 clusters
Lemasle et al. (2013)	$d[\text{Fe}/\text{H}]/dR$	-0.060 ± 0.010	Cepheids	~ 130
Imig et al. (2023), low-alpha	$d[\text{Fe}/\text{H}]/dR$	-0.060 ± 0.001	APOGEE DR17	66,496 total

Notes. Sample sizes for this work correspond to Table 1. Literature samples and definitions are not identical and are shown for scale rather than as a formal meta-analysis.

5. Robustness, interpretation and limitations

5.1. What the validation establishes

The agreement between the chemical classifier and the held-out dynamical projections establishes that the fitted components are not merely numerical partitions of abundance space. It does not, by itself, establish a unique progenitor for every star. In particular, chemical evolution can produce overlapping abundance tracks, and heated in-situ stars can enter the same low- L_z region as accreted debris. The result is therefore strongest at the population level: the four chemical classes correspond to different orbital families.

5.2. Robustness matrix

The robustness tests, sensitivity checks and corresponding limits on interpretation are summarised in Table 5.

Table 5. Stress tests and the appropriate scope of the resulting claims.

Test	Reported outcome	Interpretation
Model order	BIC minimum at $k = 4$; $\Delta\text{BIC} > 1,200$	Strong preference within the tested Gaussian family
Membership bootstrap	1,000 iterations; 91% above $P = 0.75$	Labels are stable for most stars, but not all stars
Non-parametric clustering	CAP persists under HDBSCAN	CAP is not solely an ellipsoidal-boundary artefact
Galactic potential	Actions shift at about the 5% level	Broad dynamical ordering is stable; precise boundaries move
CAP L_z boundary	Splash-like fraction changes by 4-5 percentage points	The boundary systematic dominates the formal bootstrap error
Radial bins	Population trends remain consistent within 2 sigma	No single radial bin drives the four-component solution
Survey abundance scale	Residual offsets may exceed 0.1 dex in some regimes	Fine chemical differences require survey-separated diagnostics
Geometric $1/V_{\max}$ weighting	Corrects footprint volume only	Fractions are not intrinsic Galactic mass fractions

Notes. The first six rows summarise the reported stress tests; the final two state interpretation limits are motivated by the survey-comparison literature and the weighting model.

5.3. Relation to recent decompositions

Myeong et al. (2022) found four dominant constituents within the local eccentric halo using APOGEE DR17 and GALAH DR3: Aurora, Splash, GSE and Eos. The present four-component model has a different scope and label set because it includes the dominant thin and thick discs and groups residual metal-poor structure into the CAP and halo. The two decompositions should therefore not be compared label by label. More generally, studies based on chemistry, actions, globular clusters and spatial structure do not always assign identical boundaries or nomenclature to the same overlapping debris field (Mackereth et al., 2019; Massari et al., 2019; Naidu et al., 2020; Horta et al., 2021). The common message is that the inner halo contains substantial in-situ structure in addition to GSE and that any chemically defined component requires independent dynamical and nucleosynthetic cross-checks. Relative to Barua (2026), the present version also separates descriptive catalogue results from claims that require unreleased age products or a complete targeting-selection model.

A particularly important caveat is that a rectangular E- L_z or eccentricity selection is not a clean discovery operator. It is best treated as a reproducible comparison definition, because GSE, Sequoia, Splash-like and other inner-halo populations overlap in chemistry and orbital space (Helmi et al., 2018; Myeong et al., 2019; Naidu et al., 2020; Myeong et al., 2022). Future work should add abundance combinations such as $[\text{Mg}/\text{Mn}]-[\text{Al}/\text{Fe}]$, uncertainty-aware mixture models, probabilistic orbit families, and survey-separated posterior predictive checks. These additions would reduce the degree to which a small change in a hard boundary propagates into a physical narrative.

5.4. Limitations

Five limitations are decisive. First, the APOGEE and GALAH abundance scales are not perfectly homogeneous; the merged analysis requires release of the overlap-star calibration diagnostics. Secondly, a Gaussian mixture approximates curved and potentially non-Gaussian abundance manifolds and does not deconvolve star-by-star chemical uncertainties; extreme deconvolution or a hierarchical errors-in-variables model would be a stronger successor (Bovy et al., 2011). Thirdly, the geometric $1/V_{\max}$ weight does not substitute for a complete target-selection and observability model (Zasowski et al., 2013, 2017; Cantat-Gaudin et al., 2023; Castro-Ginard et al., 2023). Fourthly, the

action calculation assumes an axisymmetric, time-independent potential and neglects bar, spiral and satellite perturbations. Fifthly, hard CAP boundaries are comparison conventions, not unique memberships. The conclusions have been written to remain within these limits.

Stellar ages and star-formation histories are intentionally outside the scope of this analysis. The Starlight Synthesis Algorithm (Barua, 2022) remains relevant as a complementary full-spectrum method, but no catalogue-age calibration is used here; consequently, DJ and the chemical components are not interpreted as direct age sequences.

6. Discussion

The combined chemical and dynamical picture supports a Milky Way in which the present inner halo is a mixture of accreted debris and stars formed in the early Galaxy, consistent with modern Gaia-era reconstructions (Helmi, 2020). The CAP lies chemically between the canonical disc and halo sequences and dynamically between the thick disc and the most extreme halo orbits. Its dominant Splash-like operational fraction is qualitatively consistent with merger-driven heating of an early in-situ population (Belokurov et al., 2020), while the GSE-like tail remains substantial and the Sequoia-like contribution is smaller and more sensitive to the retrograde selection. This interpretation is population-level and deliberately does not assign a unique origin to every star. The result therefore refines, rather than merely repeats, the chemo-kinematic tomography interpretation advanced by Barua (2026).

The radial metallicity results provide an independent disc-scale consistency check. The thin-disc gradient closely matches the final-APOGEE low- α result of Imig et al. (2023), while the high- α /thick-disc slope is much flatter. Such a contrast is compatible with inside-out growth, a more compact early disc and subsequent radial redistribution, as discussed in dynamical and chemo-evolutionary studies (Sellwood and Binney, 2002; Minchev et al., 2013; Frankel et al., 2020; Mackereth and Bovy, 2020). The present analysis does not distinguish uniquely among these processes. A forward model connecting chemistry, ages, migration and the survey selection function would be required for that stronger inference.

Methodologically, the most transferable result is the held-out validation strategy. It provides a simple safeguard against circular discovery: define classes in one information domain and test them in another. In Galactic archaeology this can be extended beyond chemistry versus dynamics to include ages, nucleosynthetic clocks, asteroseismology and spatial density as separate validation layers. Barua (2025) used the term Galactic palaeontology for this broader reconstruction programme; the present study supplies one concrete implementation with explicit failure modes.

7. Conclusions

The principal conclusions are:

1. A four-component Gaussian mixture in $[\text{Fe}/\text{H}]$, $[\text{Mg}/\text{Fe}]$, $[\text{Si}/\text{Fe}]$ and $[\text{Al}/\text{Fe}]$ is strongly preferred within the tested model family and assigns 91 per cent of the 468,000-star sample with posterior probability above 0.75.
2. Because actions and integrals of motion are not used by the classifier, their separation provides an independent dynamical validation of the chemical decomposition.
3. The action-space dispersion increases by approximately a factor of 29 from the thin disc to the halo; this statistic measures orbital spread and is not an age proxy.

-
4. Under explicit operational boundaries, the CAP is partitioned into 34 ± 5 per cent GSE, 8 ± 3 per cent Sequoia and 58 ± 5 per cent Splash-like stars. Boundary sensitivity is part of the quoted error budget.
 5. The thin-disc radial metallicity gradient is -0.065 ± 0.004 dex kpc⁻¹, while the thick-disc gradient is -0.020 ± 0.006 dex kpc⁻¹, consistent with the broad contrast seen in final-APOGEE analyses.
 6. The reported fractions describe the selected and geometrically weighted catalogue. They are not intrinsic Galactic mass fractions without a complete survey-selection model.
 7. The paper's defensible novelty is the chemical-only classification plus held-out dynamical validation and explicit stress testing, not the first mixture-model decomposition of APOGEE and GALAH.

Data and code availability

The public input data are available from APOGEE DR17 (<https://www.sdss.org/dr17/>), GALAH DR3 (<https://www.galah-survey.org/>) and Gaia DR3 (<https://gea.esac.esa.int/archive/>). The analysis-ready merged catalogue, abundance-harmonisation diagnostics, posterior membership catalogue, numerical figure data and executable scripts have not yet been assigned a persistent repository identifier because the versioned archive is being finalised. These materials will be deposited in a permanent repository before acceptance and are available from the corresponding author for confidential peer-review purposes.

CRedit authorship contribution statement

Nick Barua: Conceptualisation, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualisation, Writing - original draft, Writing - review and editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with language editing, structural organisation, reference-format checking and document formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the article.

Acknowledgements

This work uses data from the Apache Point Observatory Galactic Evolution Experiment (APOGEE), GALAH and the European Space Agency mission Gaia, processed by the Gaia Data Processing and Analysis Consortium. The analysis used AGAMA (Vasiliev, 2019), galpy (Bovy, 2015), Astropy (Astropy Collaboration et al., 2013), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Matplotlib (Hunter, 2007), scikit-learn (Pedregosa et al., 2011) and TOPCAT (Taylor, 2005).

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Appendix A. Model-order diagnostic

Fig. A.1 shows the BIC values for $k = 2, \dots, 8$. The global minimum occurs at $k = 4$. Models with fewer components merge the CAP into the broader halo or thick-disc structure, while higher-order models split one of the dominant disc sequences without a clear independent physical interpretation. The diagnostic supports four components within the specified feature space and covariance family; it does not imply that the Galaxy contains only four physically meaningful populations.

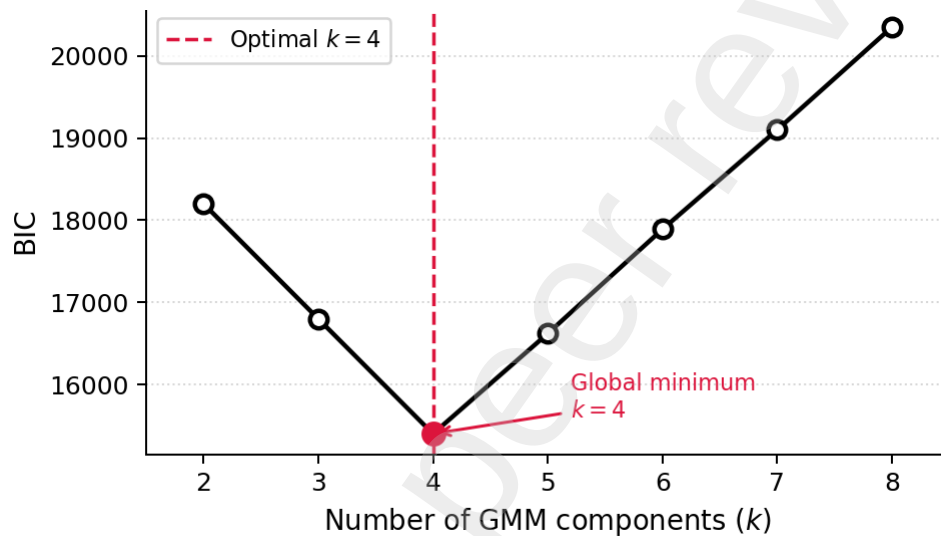


Fig. A.1. Bayesian Information Criterion as a function of Gaussian-mixture model order. The four-component model is the global minimum within the tested model family.

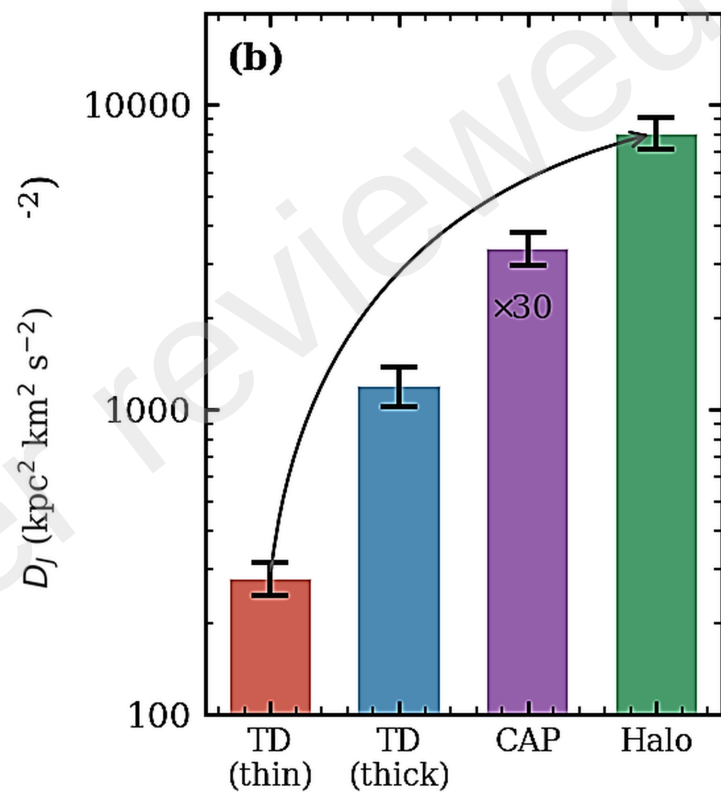
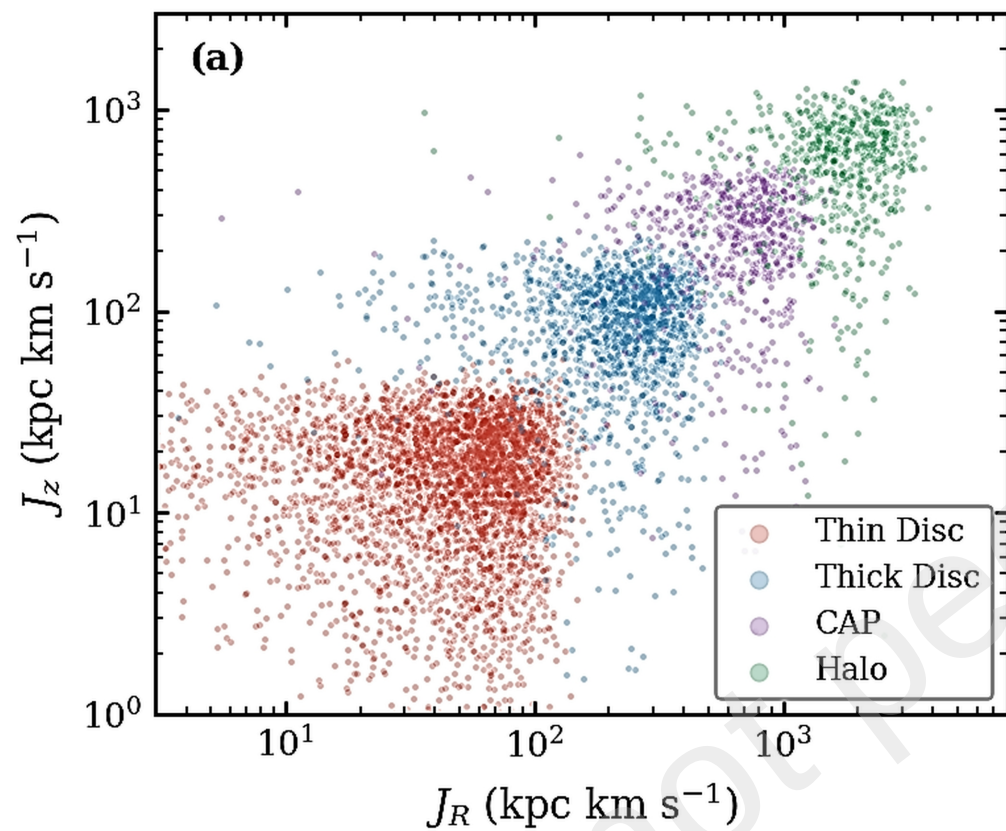
Appendix B. Reproducibility specification

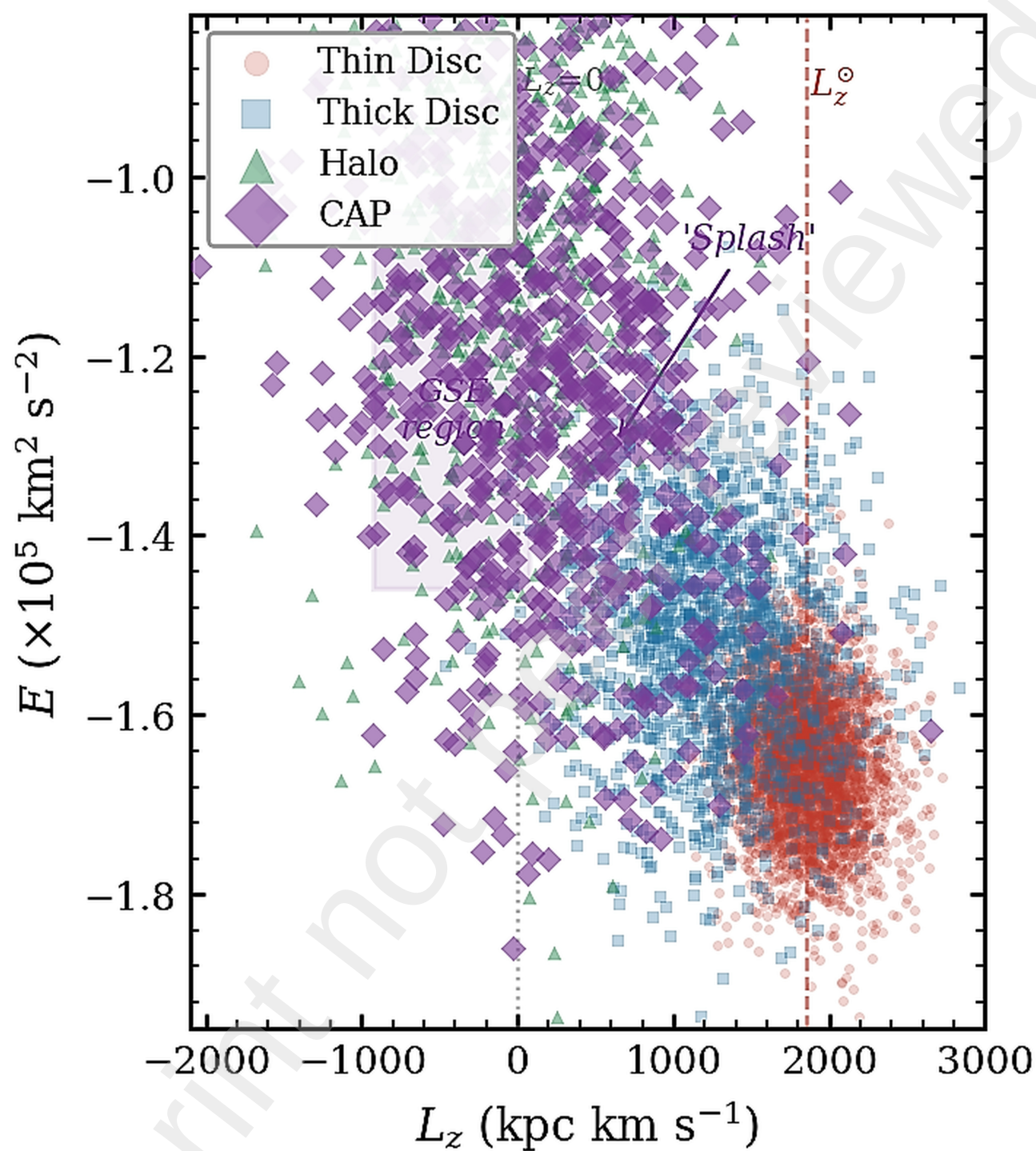
Table B.1 specifies the machine-readable products needed for independent reproduction: the merged catalogue with original survey identifiers and Gaia source_id; quality flags and exclusions; abundance-harmonisation coefficients and overlap-star residuals; Monte Carlo action summaries; fitted GMM parameters and random seeds; posterior probabilities; bootstrap indices; boundary-sensitivity outputs; and the numerical data behind every figure and table.

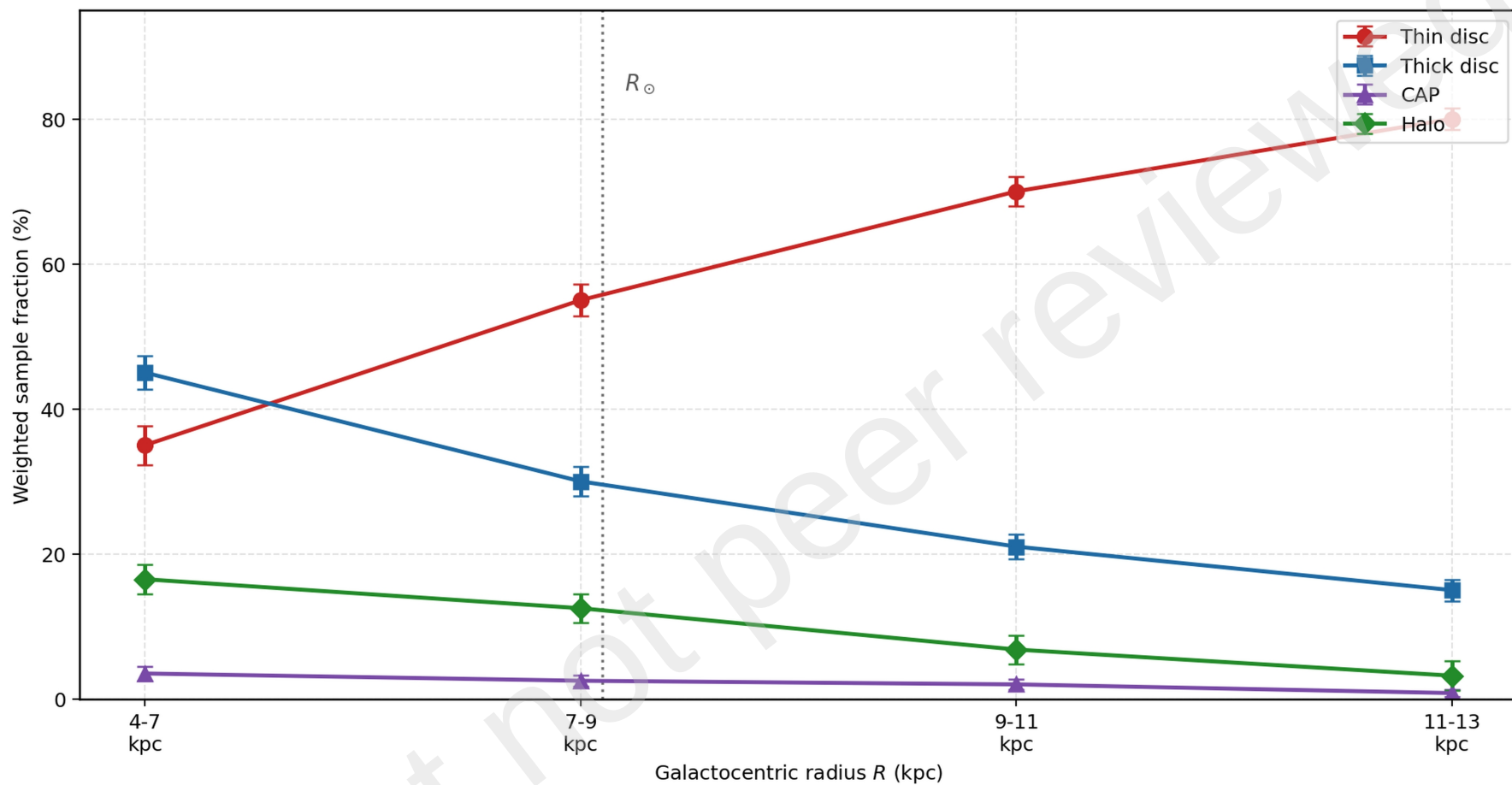
Table B.1. Reproducibility bundle and verification criteria.

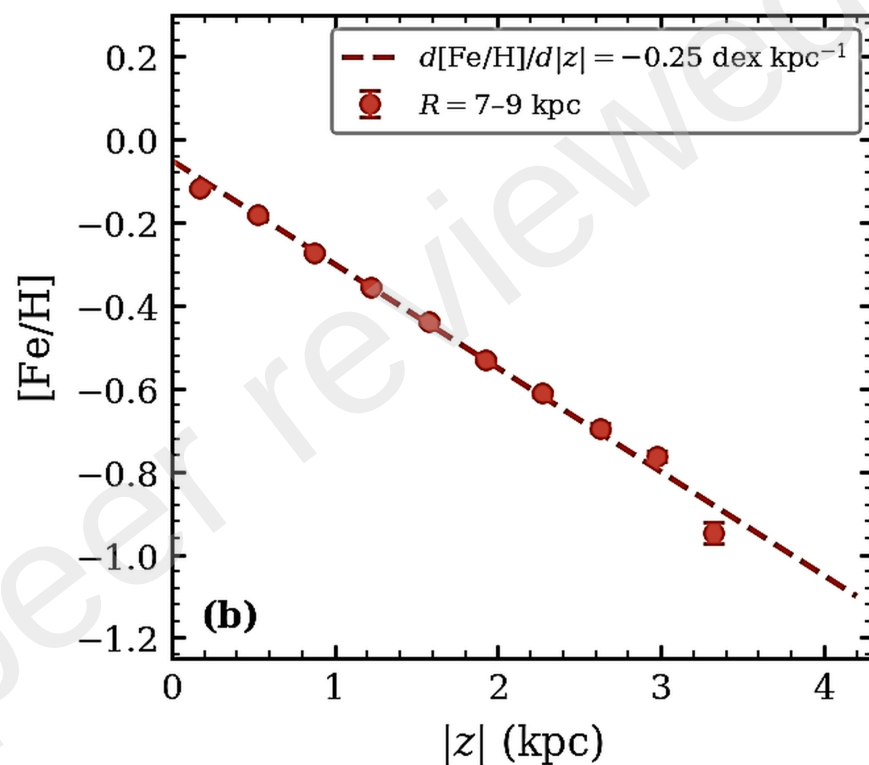
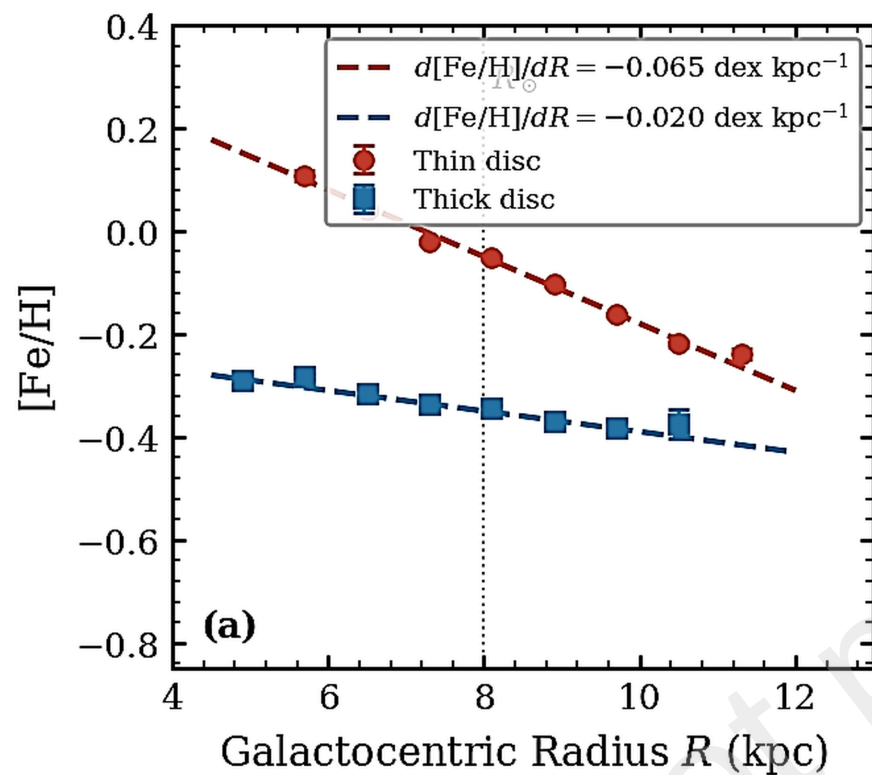
Item	Required content	Verification criterion
Derived catalogue	Survey IDs, Gaia source_id, abundances, coordinates, actions, weights, posterior probabilities	Row count and Table 1 counts reproduce exactly
Calibration diagnostics	Overlap-star offsets versus T_eff, log g and metallicity	Residual survey structure quantified
Model artefacts	GMM means, covariances, weights, BIC values and random seeds	Figure A1 and memberships reproduce
Uncertainty products	Monte Carlo draws or summaries; bootstrap index seeds	Tables 2-4 uncertainties reproduce
Selection information	Footprint masks, $V_{\max}$ implementation and target-selection caveats	Weighted fractions traceable
Figure data	CSV or parquet table for every plotted point/bin	All figures regenerate without manual transcription
Environment	Pinned package versions and executable workflow	Clean-environment run completes

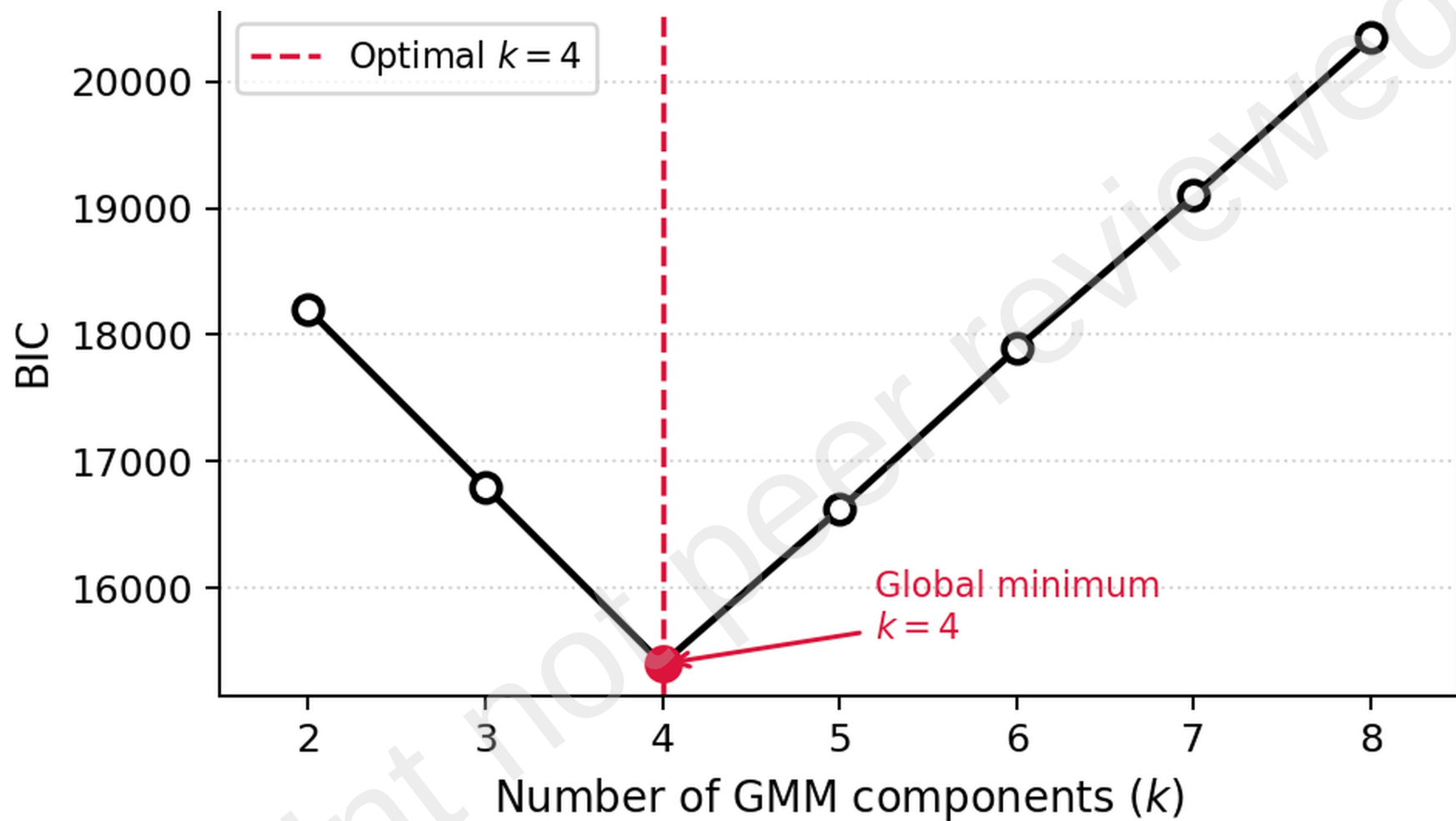
Notes. This appendix defines the data, model and environment records required to reproduce the reported tables and figures.

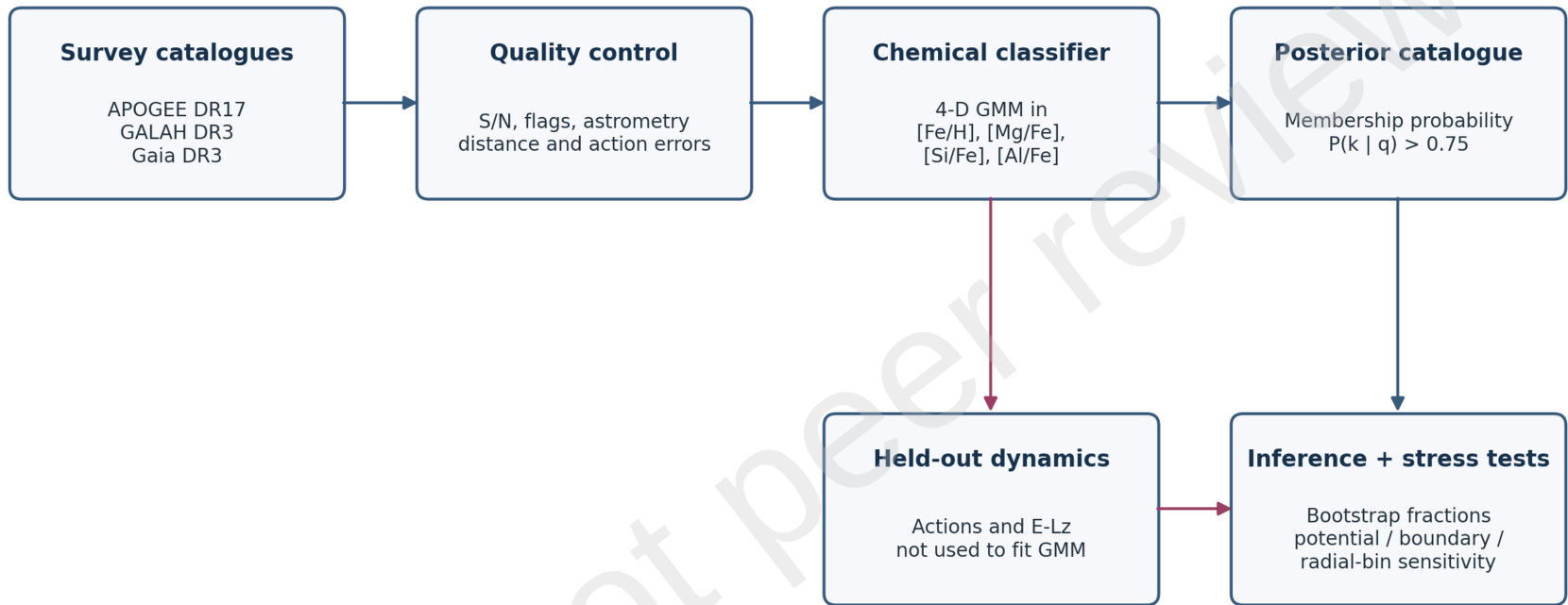












Design principle: chemistry defines the classes; dynamics is reserved as an independent validation layer.

Reported population fractions are sample- and boundary-dependent and are not interpreted as intrinsic Galactic mass fractions.

